

Signed binomial supercongruences

A proof of the conjecture recorded in OEIS A352373

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Abstract

The OEIS entry A352373 records a conjecture that every sequence $a_{r,s}(n) = [x^n](1+x)^{rn}(1-x)^{sn}$, with arbitrary integer r, s , satisfies the two-term congruences modulo p^{3h} for primes $p \geq 5$. We give a proof, allowing the coefficient index to be any nonnegative integer multiple tn . The argument separates indices divisible by p from the remaining indices. Jacobsthal's binomial congruence treats the first part; a multiscale alternating reciprocal-square cancellation treats the second. The same proof gives modulus 3^{3h-1} at $p = 3$. We derive four OEIS specializations, a denominator restriction for a Möbius transform, and formal Lambert-series and trilogarithm expansions. Exact computations check 13,690 family congruences and 15,154 proof-component instances. The proof uses a precisely cited classical congruence; it has not been independently refereed or formally verified, and publication priority is not claimed.

Main result. For $r, s \in \mathbb{Z}$, $t \in \mathbb{Z}_{\geq 0}$, and $a(n) = [x^{tn}](1+x)^{rn}(1-x)^{sn}$,

$$a(mp^h) \equiv a(mp^{h-1}) \pmod{p^{3h}} \quad (p \geq 5).$$

Here $m, h \geq 1$ are arbitrary integers; in particular, $p \nmid m$ is *not* required. Negative exponents are interpreted as formal power series. The conjecture in A352373 is exactly the case $t = 1$.

1 The selected problem and the theorem

The entry [A352373](#), authored by Peter Bala, records both an individual supercongruence conjecture and its two-parameter extension [1]. The individual sequence is

$$a(n) = [x^n] \frac{1}{(1-x)^{2n}(1-x^2)^n} = [x^n](1+x)^{-n}(1-x)^{-3n}. \quad (1.1)$$

The general assertion concerns the same congruence for every integer pair (r, s) in $[x^n](1+x)^{rn}(1-x)^{sn}$. In the entry retrieved for this note, that assertion is still explicitly marked as a conjecture.

Rather than finding a counterexample, the exact tests suggested a uniform proof. We work with the larger family

$$a_{r,s,t}(n) := [x^{tn}](1+x)^{rn}(1-x)^{sn}, \quad r, s \in \mathbb{Z}, \quad t, n \in \mathbb{Z}_{\geq 0}. \quad (1.2)$$

All of these coefficients are integers, including when r or s is negative. Put

$$\delta_p = \begin{cases} 1, & p = 3, \\ 0, & p \geq 5, \end{cases} \quad \text{for odd primes } p. \quad (1.3)$$

Theorem 1.1. *For every odd prime p , all integers r, s , all nonnegative integers t , and all positive integers m, h ,*

$$a_{r,s,t}(mp^h) \equiv a_{r,s,t}(mp^{h-1}) \pmod{p^{3h-\delta_p}}. \quad (1.4)$$

A homogeneous coefficient statement, proved in [section 4](#), implies this theorem immediately. The proof uses the classical Jacobsthal congruence in the version allowing negative integer upper entries [[7](#), Lemma 5.1]. Its organization follows the established binomial-descent and block-cancellation approach to supercongruences [[6](#), Section 2]. The purpose here is the explicit application to the whole coefficient family, not a claim that those proof techniques are new.

The distinction between a conjectural entry and a new theorem matters: [A234839](#), the case $(r, s, t) = (1, 2, 1)$, already has this supercongruence, as recorded in the entry and in [[6](#), Example 3.3]. We do not count that special case as a new result [[5](#)]. The limited literature check underlying the present note is described in [section 9](#) and in the accompanying source log.

2 Elementary preliminaries and the classical input

For an integer u and a nonnegative integer k , use

$$\binom{u}{k} = \frac{u(u-1)\cdots(u-k+1)}{k!}, \quad \binom{u}{0} = 1.$$

In particular, for $v \geq 1$,

$$\binom{-v}{k} = (-1)^k \binom{v+k-1}{k} \in \mathbb{Z}. \quad (2.1)$$

If $u \geq 0$ and $k > u$, the coefficient is zero. We never need binomial coefficients with negative lower entries in the proof.

Let $\mathbb{Z}_{(p)}$ be the ring of rational numbers whose reduced denominators are not divisible by p . A congruence in this ring means divisibility by the indicated power of p inside $\mathbb{Z}_{(p)}$. Thus all reciprocal squares $1/j^2$ with $p \nmid j$ are legitimate. We set $v_p(0) = +\infty$. No convergence of infinite p -adic series will be needed.

Lemma 2.1 (One-digit binomial descent). *Let p be any prime, $h \geq 1$, $p^h \mid U$, and $k \geq 0$. Then*

$$\binom{U-1}{k} \equiv (-1)^{k-\lfloor k/p \rfloor} \binom{U/p-1}{\lfloor k/p \rfloor} \pmod{p^h}. \quad (2.2)$$

Proof. Splitting the product into indices divisible and not divisible by p gives the exact identity in $\mathbb{Q}$

$$\binom{U-1}{k} = \binom{U/p-1}{\lfloor k/p \rfloor} \prod_{\substack{1 \leq i \leq k \\ p \nmid i}} \frac{U-i}{i}.$$

Each factor in the product is congruent to -1 modulo p^h in $\mathbb{Z}_{(p)}$. There are $k - \lfloor k/p \rfloor$ such factors. The remaining binomial coefficient is an integer, even for negative U , so multiplication preserves the congruence. This argument also allows a vanishing remaining coefficient. $\square$

This standard descent identity also appears as [[6](#), Lemma 2.4].

Lemma 2.2 (Jacobsthal; imported classical theorem). *Let p be an odd prime, let X, Y be integers with $Y \geq 0$ and $p \mid X, Y$, and suppose $\binom{X/p}{Y/p} \neq 0$. If $XY(X - Y) \neq 0$, then*

$$\frac{\binom{X}{Y}}{\binom{X/p}{Y/p}} - 1 \in p^{v_p(XY(X-Y)) - \delta_p} \mathbb{Z}_{(p)}. \quad (2.3)$$

In the remaining nonvanishing cases $Y = 0$ or $X = Y$, the ratio equals one.

This is the odd-prime part of [7, Lemma 5.1]: the exponent there is $v_p(XY(X - Y)/12)$. Its loss is one at $p = 3$ and zero at $p \geq 5$. That reference includes negative upper entries. The familiar $p \geq 5$ valuation form is also stated in [6, Lemma 2.1]. We invoke this known theorem, rather than claim a new proof of it.

There is no use of a ratio $0/0$. If $\binom{X/p}{Y/p} = 0$, then $X \geq 0$ and $Y > X$, and $\binom{X}{Y} = 0$ as well. Conversely, a zero fine coefficient has a zero coarse coefficient. These support conditions are unchanged by division of both entries by p .

Lemma 2.3 (Alternating reciprocal-square blocks). *Let p be an odd prime, $\ell \geq 1$, and $q \geq 0$. Then*

$$H_{\ell,q} := \sum_{\substack{qp^\ell \leq j < (q+1)p^\ell \\ p \nmid j}} \frac{(-1)^j}{j^2} \equiv 0 \pmod{p^\ell}. \quad (2.4)$$

Proof. Write $P = p^\ell$. Within the block, pair $j = qP + a$ with $j' = qP + P - a$, for $1 \leq a < P$ and $p \nmid a$. The pairing has no fixed point since P is odd. We have $j' \equiv -j \pmod{P}$ and $(-1)^{j'} = -(-1)^j$. Their reciprocal-square contributions cancel modulo P . All denominators involved are units modulo P . $\square$

3 The weighted cancellation that supplies the third power

The difficult part is not the unweighted congruence in lemma 2.3, but retaining it in the presence of binomial weights. The following lemma makes the necessary compatibility explicit. Its telescoping step is a finite-block version of the weighted congruence method; compare [7, Lemma 5.6].

Lemma 3.1 (Complementary-weight cancellation). *Let p be an odd prime, $h \geq 1$, and $A, B, K \in \mathbb{Z}$ with $K > 0$ and $p^h \mid A, B, K$. Then*

$$W := \sum_{\substack{1 \leq j < K \\ p \nmid j}} \frac{(-1)^j}{j^2} \binom{A-1}{j-1} \binom{B-1}{K-j-1} \equiv 0 \pmod{p^h}. \quad (3.1)$$

Proof. For $1 \leq \ell \leq h$ and $0 \leq q < K/p^\ell$, define integer weights

$$U_\ell(q) = \binom{A/p^\ell - 1}{q} \binom{B/p^\ell - 1}{K/p^\ell - 1 - q}. \quad (3.2)$$

All lower entries are nonnegative; negative upper entries cause no integrality problem by (2.1).

The first descent has a minus sign. For $p \nmid j$ write $j = pq + a$ with $1 \leq a < p$. Because $p \mid K$,

$$\lfloor (j-1)/p \rfloor = q, \quad \lfloor (K-j-1)/p \rfloor = K/p - 1 - q.$$

Applying [lemma 2.1](#) to both binomial coefficients, modulo p^h , produces a sign with exponent

$$(j-1) + (K-j-1) - (q + (K/p - 1 - q)) = K - K/p - 1.$$

The difference $K - K/p = (p-1)K/p$ is even. Consequently,

$$\binom{A-1}{j-1} \binom{B-1}{K-j-1} \equiv -U_1(\lfloor j/p \rfloor) \pmod{p^h}. \quad (3.3)$$

Introduce the sums

$$T_\ell = \sum_{\substack{1 \leq j < K \\ p \nmid j}} \frac{(-1)^j}{j^2} U_\ell(\lfloor j/p^\ell \rfloor). \quad (3.4)$$

Equation (3.3) says $W \equiv -T_1 \pmod{p^h}$.

Every subsequent descent has a plus sign. Suppose $1 \leq \ell < h$ and put $L = K/p^\ell$. Since $p \mid L$,

$$\lfloor q/p \rfloor + \lfloor (L-1-q)/p \rfloor = L/p - 1.$$

Both upper parameters A/p^ℓ and B/p^ℓ are divisible by $p^{h-\ell}$. Another application of [lemma 2.1](#) gives

$$U_\ell(q) \equiv U_{\ell+1}(\lfloor q/p \rfloor) \pmod{p^{h-\ell}}. \quad (3.5)$$

Indeed, the combined sign exponent is $(L-1) - (L/p - 1) = L - L/p$, which is even. This is why the complementary lower entries in (3.2) are useful.

Block cancellation compensates for the declining precision. Group the difference $T_\ell - T_{\ell+1}$ by $q = \lfloor j/p^\ell \rfloor$. Since K is a multiple of p^ℓ , all blocks are complete, and

$$T_\ell - T_{\ell+1} = \sum_{q=0}^{K/p^\ell-1} (U_\ell(q) - U_{\ell+1}(\lfloor q/p \rfloor)) H_{\ell,q}. \quad (3.6)$$

The bracket belongs to $p^{h-\ell}\mathbb{Z}$ by (3.5), while $H_{\ell,q} \in p^\ell\mathbb{Z}_{(p)}$ by [lemma 2.3](#). Every product in (3.6) therefore belongs to $p^h\mathbb{Z}_{(p)}$.

At the last level, each weight is constant on a block of length p^h :

$$T_h = \sum_{q=0}^{K/p^h-1} U_h(q) H_{h,q} \equiv 0 \pmod{p^h}.$$

Thus $T_1 \equiv T_2 \equiv \dots \equiv T_h \equiv 0 \pmod{p^h}$, and $W \equiv 0 \pmod{p^h}$ follows. If $h = 1$, only the first and last steps are needed. $\square$

Remark 3.2. The proof does not replace a weight by a constant modulo p^h across an entire block of length p^h in a single step. At level ℓ it has only $p^{h-\ell}$ precision; the block sum supplies exactly the missing p^ℓ . Ignoring this precision bookkeeping would leave a gap.

4 Proof of the coefficient supercongruence

Define, for $A, B \in \mathbb{Z}$ and $K \geq 0$,

$$C(A, B; K) := [x^K](1-x)^A(1+x)^B = \sum_{j=0}^K (-1)^j \binom{A}{j} \binom{B}{K-j}. \quad (4.1)$$

The finite convolution is valid for negative as well as positive exponents in the ring of formal power series.

Lemma 4.1 (Descent of the multiple indices). *Let p be an odd prime, $h \geq 1$, $K \geq 0$, and $p^h \mid A, B, K$. For $0 \leq j \leq K$ with $p \mid j$,*

$$\binom{A}{j} \binom{B}{K-j} \equiv \binom{A/p}{j/p} \binom{B/p}{(K-j)/p} \pmod{p^{3h-\delta_p}}. \quad (4.2)$$

Proof. If one coarse binomial coefficient is zero, its fine counterpart is zero as well, so both products vanish. Assume otherwise. Put

$$e = \min\{h, v_p(j)\}.$$

This is an integer between 1 and h , with $e = h$ when $j = 0$. Both j and $K - j$ are divisible by p^e . Applying lemma 2.2 to each factor gives

$$\binom{A}{j} \binom{B}{K-j} = \binom{A/p}{j/p} \binom{B/p}{(K-j)/p} (1 + E), \quad E \in p^{h+2e-\delta_p} \mathbb{Z}_{(p)}. \quad (4.3)$$

For example, $v_p(Aj(A-j)) \geq h + 2e$ whenever the three factors are nonzero. Endpoint ratios equal one, so they satisfy the same error bound without invoking a valuation of a nonzero product.

If $e < h$, then $v_p(j) = v_p(K-j) = e$, since $p^h \mid K$. The identity

$$\binom{u}{v} = \frac{u}{v} \binom{u-1}{v-1} \quad (v > 0)$$

shows that each coarse coefficient in (4.3) is divisible by p^{h-e} . Its upper entry is divisible by p^{h-1} , and its lower entry has valuation $e-1$. If $e = h$, the same conclusion means only that the coefficients are integers. Their product is therefore always divisible by $p^{2(h-e)}$.

Subtracting the coarse product from (4.3), the valuation of the difference is at least

$$(h + 2e - \delta_p) + 2(h - e) = 3h - \delta_p.$$

Since the difference is an integer, this proves the claimed ordinary integer congruence. $\square$

Theorem 4.2 (Homogeneous coefficient theorem). *For an odd prime p , $h \geq 1$, and integers A, B, K with $K \geq 0$ and $p^h \mid A, B, K$,*

$$C(A, B; K) \equiv C(A/p, B/p; K/p) \pmod{p^{3h-\delta_p}}. \quad (4.4)$$

Proof. For $K = 0$ both sides are one. Assume $K > 0$ and split (4.1) into the terms $p \mid j$ and $p \nmid j$. For the first part, lemma 4.1 applies term by term. Because p is odd, $(-1)^j = (-1)^{j/p}$ whenever $p \mid j$. Consequently,

$$\sum_{\substack{0 \leq j \leq K \\ p \mid j}} (-1)^j \binom{A}{j} \binom{B}{K-j} \equiv C(A/p, B/p; K/p) \pmod{p^{3h-\delta_p}}. \quad (4.5)$$

For the second part, both j and $K - j$ are units in $\mathbb{Z}_{(p)}$. The exact binomial identity yields

$$\binom{A}{j} \binom{B}{K-j} = \frac{AB}{j(K-j)} \binom{A-1}{j-1} \binom{B-1}{K-j-1}. \quad (4.6)$$

Also,

$$\frac{1}{j(K-j)} + \frac{1}{j^2} = \frac{K}{j^2(K-j)} \in p^h \mathbb{Z}_{(p)}.$$

As $AB \in p^{2h} \mathbb{Z}$, summing (4.6) over the unit indices with signs gives

$$\sum_{\substack{1 \leq j < K \\ p \nmid j}} (-1)^j \binom{A}{j} \binom{B}{K-j} \equiv -ABW \equiv 0 \pmod{p^{3h}}, \quad (4.7)$$

where the last congruence is lemma 3.1. If A or B is zero, (4.6) remains valid and the entire sum is zero. Combining (4.5) and (4.7) proves (4.4). $\square$

Proof of theorem 1.1. Take $A = sm p^h$, $B = rm p^h$, and $K = tm p^h$ in theorem 4.2. By definition, $C(A, B; K) = a_{r,s,t}(m p^h)$, and its scaled counterpart is $a_{r,s,t}(m p^{h-1})$. $\square$

Remark 4.3 (Additional common divisibility). If $m = p^v m_0$ with $p \nmid m_0$, the same theorem at height $h + v$ gives the stronger modulus $p^{3(h+v)-\delta_p}$. More generally, theorem 4.2 permits any common power of p dividing all three parameters A, B, K .

A concrete check illustrates why cancellation is indispensable. For A352373 at $n = 5$, the local parameters are $(A, B, K) = (-15, -5, 5)$. The $j = 1$ summand is $1050 = 42 \cdot 5^2$, not a multiple of 5^3 . The sum of the four nonmultiple-index terms is $-8250 = -66 \cdot 5^3$. Thus the extra power comes from their sum, not from an unjustified termwise estimate.

5 Consequences for the selected OEIS entries

The following substitutions are immediate from $1 - x^2 = (1 - x)(1 + x)$. Each of these four entries explicitly records the p^{3h} congruence as conjectural in the version consulted for this note [1, 2, 3, 4].

Sequence	r	s	t	Defining coefficient
A352373	-1	-3	1	$[x^n](1-x)^{-2n}(1-x^2)^{-n}$
A348410	-1	-2	1	$[x^n](1-x)^{-n}(1-x^2)^{-n}$
A351856	-1	-2	2	$[x^{2n}](1-x)^{-n}(1-x^2)^{-n}$
A351857	-2	-4	1	$[x^n](1-x)^{-2n}(1-x^2)^{-2n}$

The coefficient-index extension $t \geq 0$ is particularly useful for A351856; its displayed coefficient representation requires $t = 2$, beyond the index used in the OEIS family conjecture.

For the selected sequence (1.1), expanding the two negative-binomial factors also gives the positive sum

$$a(n) = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{3n-2k-1}{n-2k} \binom{n+k-1}{k}, \quad n \geq 1. \quad (5.1)$$

For instance $a(1) = 2$, $a(3) = 74$, $a(5) = 3252$, and $a(7) = 154352$. The theorem proves both

$$a(m p^h) \equiv a(m p^{h-1}) \pmod{p^{3h}} \quad (p \geq 5)$$

and the additional relation

$$a(m 3^h) \equiv a(m 3^{h-1}) \pmod{3^{3h-1}}.$$

The latter assertion is a consequence of the weaker classical Jacobsthal exponent at $p = 3$, not an inference from finite examples.

6 A generating-function consequence

For any fixed (r, s, t) let $a_n = a_{r,s,t}(n)$ and define

$$b_n = \frac{1}{n^3} \sum_{d|n} \mu(d) a_{n/d}, \quad n \geq 1, \quad (6.1)$$

where μ is the Möbius function.

Proposition 6.1. *For every $n \geq 1$,*

$$b_n \in \mathbb{Z}[1/6], \quad \text{and more precisely} \quad 3b_n \in \mathbb{Z}[1/2]. \quad (6.2)$$

In particular, b_n is an integer when $\gcd(n, 6) = 1$. The following identities hold as formal power series over $\mathbb{Q}$:

$$\sum_{n \geq 1} a_n z^n = \sum_{n \geq 1} n^3 b_n \frac{z^n}{1 - z^n}, \quad (6.3)$$

$$\sum_{n \geq 1} \frac{a_n}{n^3} z^n = \sum_{n \geq 1} b_n \operatorname{Li}_3(z^n), \quad \operatorname{Li}_3(z) = \sum_{k \geq 1} \frac{z^k}{k^3}. \quad (6.4)$$

Proof. Fix an odd prime divisor p of n and write $n = p^h m$ with $p \nmid m$. Terms for which $p^2 \mid d$ vanish in the Möbius sum, so its numerator equals

$$\sum_{e|m} \mu(e) \left(a_{(m/e)p^h} - a_{(m/e)p^{h-1}} \right).$$

By [theorem 1.1](#), this is divisible by $p^{3h-\delta_p}$. Dividing by n^3 proves $v_p(b_n) \geq 0$ for $p \geq 5$, and $v_3(b_n) \geq -1$. The reduced denominator of b_n divides n^3 , so no other odd primes can occur. This proves (6.2) and the assertion for $\gcd(n, 6) = 1$.

Möbius inversion of (6.1) gives $a_n = \sum_{d|n} d^3 b_d$. The coefficient of z^n on the right of (6.3) is precisely this divisor sum. The coefficient on the right of (6.4) is n^{-3} times the same divisor sum. For each degree only finitely many divisors contribute, so no analytic convergence assumption is required. $\square$

For [A352373](#), the first seven b_n are

$$2, \quad \frac{5}{4}, \quad \frac{8}{3}, \quad \frac{59}{8}, \quad 26, \quad \frac{308}{3}, \quad 450.$$

This illustrates both the integrality away from 2, 3 and the genuine possibility of a factor 3 in the denominator. The transformation itself is ordinary Möbius inversion; the content supplied by the theorem is its denominator restriction.

7 Sharpness and failed generalizations

The chosen conjecture survives; the following are counterexamples only to *stronger or broader* statements.

The uniform exponent cannot be increased. For the selected sequence,

$$a(5) - a(1) = 3252 - 2 = 3250 = 26 \cdot 5^3.$$

It is not divisible by 5^4 . Thus a universal modulus p^{3h+1} , and a fortiori p^{4h} , is false even within A352373. This example establishes uniform sharpness; it does not assert that equality of valuations holds for every prime or every height.

The full exponent cannot be extended to $p = 3$ or $p = 2$. The same sequence gives

$$a(3) - a(1) = 72 = 8 \cdot 3^2, \quad a(2) - a(1) = 10.$$

The first is not divisible by 3^3 and shows that the one-power loss at $p = 3$ is necessary in a uniform statement. The second is not divisible by 2^3 . No general 2-adic strengthening is claimed here.

An arbitrary integer polynomial does not suffice. Consider instead $c(n) = [x^n](1 + x + x^2)^n$. Direct expansion gives $c(1) = 1$ and

$$c(5) = \binom{5}{0}\binom{0}{0} + \binom{5}{2}\binom{2}{1} + \binom{5}{4}\binom{4}{2} = 1 + 20 + 30 = 51.$$

Consequently $c(5) - c(1) = 50$ is not divisible by 5^3 . Even a palindromic quadratic with constant coefficient one need not satisfy the theorem. The opposite signs in $(1 - x)^A(1 + x)^B$ enable the specific alternating cancellation used in the proof.

8 Reproducible computation and proof audit

The archive contains a standard-library-only Python program, `code/verify.py`, together with its complete output `results/verification.json` and a row-level record `results/family_cases.csv`. Run

```
python code/verify.py --output results
```

The recorded run used Python 3.13.5. All checks passed.

Test collection	Checks
Parameter grid, primes at least five	9,295
Parameter grid, prime three with exponent $3h - 1$	4,225
Seeded additional family tests	150
Focused higher-height tests	20
Total family congruences	13,690
Individual proof-component congruences	15,154
Möbius-transform denominator checks	500

The grid uses $-6 \leq r, s \leq 6$ and $0 \leq t \leq 4$, with the explicitly listed (p, h, m) cases stored in the JSON file. The focused tests include height four at $p = 5$, height five at $p = 3$, and indices as large as $11^3 = 1331$. There are 10,050 family cases with a nonzero difference; the other 3,640 include legitimate zero or constant cases and are not hidden in the totals. Randomized tests have the fixed seed 352373.

Two independent exact evaluators are compared whenever a sequence value is computed. The first expands the binomial convolution. For the second, write $F(x) = (1 + x)^R(1 - x)^S = \sum c_k x^k$. Logarithmic differentiation and coefficient comparison give

$$(1 - x^2)F'(x) = ((R - S) - (R + S)x)F(x), \quad kc_k = (R - S)c_{k-1} + (k - 2 - R - S)c_{k-2}, \quad (8.1)$$

with $c_{-1} = 0$ and $c_0 = 1$. This uses $O(K)$ integer-arithmetic steps to obtain c_K , with varying integer sizes. Each division is checked to be exact. It is *not* a modular recurrence that divides by a possible nonunit k .

The proof-component tests include the first-descent minus sign, every subsequent weighted transition, the terminal blocks, support-zero cases, and the stronger p^{3h} vanishing of the off-multiple sum even at $p = 3$. They test arithmetic instances of the proof, not just its conclusion. They are still finite tests, not a formal verification of the universally quantified theorem.

For a further transparent check, the selected sequence gives the following valuations; full differences are recorded in the JSON file.

p	h	$v_p(a(p^h) - a(p^{h-1}))$	Guaranteed exponent
5	1	3	3
5	2	6	6
5	3	9	9
7	1	3	3
3	1	2	2
3	2	5	5

9 Status, attribution, and what has actually been established

This note supplies an all-parameter argument for [theorem 1.1](#), not merely evidence for a small case. The only non-elementary imported result is the classical congruence in [lemma 2.2](#); all subsequent deductions, including the full weighted cancellation, are given above. No unproved auxiliary conjecture is used.

The source check covered the relevant OEIS entries and the two primary papers cited for the congruence machinery. The general assertion remains labelled conjectural in A352373 as retrieved on 18 September 2026. An already-proved special case was explicitly separated from the target. This is not an exhaustive literature review, and an equivalent general result could exist in a different formulation. Accordingly, the claim is *a proof of the recorded assertion and its stated extensions*, not verified priority for a new discovery.

The document is an unrefereed AI-assisted research draft. Neither an independent mathematician's review nor a proof-assistant check was performed. The arithmetic tests are reproducible but cannot replace such review. The accompanying `PROOF_AUDIT.md` identifies the valuation, sign, boundary, and dependency checks that are especially important for independent scrutiny. The proposed OEIS note in the archive is a draft only; nothing has been submitted to OEIS.

References

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